

Eric Johnson

eric@ekjohnson.dev

Career Objective

Computer Science undergraduate with a Data Science minor and three years of experience at LANL in HPC and infrastructure engineering, plus applied research experience in machine learning applications in hardware Trojan detection. Seeking a summer 2027 undergraduate internship with CAI in applied computer science.

Technical Skills

HPC & Scientific Computing: MPI, HPL, Slurm, OpenCHAMI, parallel benchmarking, GitLab CI/CD

Machine Learning: PyTorch, Python Outlier Detection (PyOD), Hugging Face Transformers, Keras/TensorFlow, CNNs

Languages: Python, C, Java, Bash, R, SQL, YAML, PowerShell

Systems: Linux (Rocky, RHEL, Ubuntu), Docker, Podman, systemd-nspawn, Git

Experience

Los Alamos National Laboratory (LANL) Student Intern

High-Performance Computing – Supercomputer Institute Intern

May – August 2026

- Developed and presented a novel boot method leveraging containers and hypervisors to enable rapid node reprovisioning, benchmarked using MPI and HPL against traditional methods; presented findings at LANL's 2026 HPC Intern Showcase.
- Deployed and administered an HPC compute cluster with a peer team, using OpenCHAMI and Slurm for provisioning and job scheduling, leveraging Ansible and CI/CD pipelines for configuration.

Sensitive & Special Operations – IT Systems Automation Intern

June 2023 – May 2026

- Briefed LANL's Deputy Field Intelligence Element (FIE) Director on automation initiatives that reduced maintenance overhead and strengthened reliability of the team's development-environment infrastructure.
- Stood up and administered Red Hat Satellite and GitLab servers, enabling centralized Linux package distribution and source code management across group's IT development environment.
- Designed and implemented Ansible Automation Platform and PowerShell workflows for Hyper-V hypervisors, replacing manual per-VM configuration with automated, repeatable provisioning and state management.
- Authored and maintained cross-platform documentation (Linux, Windows, Ansible) and provided troubleshooting support, giving new hires self-service references for common workflows and environment setup.

Research Experience

Intel Undergraduate Research Fellow

August – December 2026

New Mexico Institute of Mining & Technology, Department of Computer Science

- Evaluating anomaly detection models from the Python Outlier Detection (PyOD) library for golden-chip free hardware Trojan detection on power and electromagnetic side channel traces from AES Trojan variants (Trust Hub / IEEE DataPort).
- Benchmarking detector effectiveness against classical baselines on unlabeled, imbalanced measurement data where dormant Trojans separate from baseline circuits only by dynamic power draw.

Education

B.S. in Computer Science | Minor in Data Science | GPA: 3.8

Expected: May 2028

New Mexico Institute of Mining & Technology – Socorro, NM

- **Relevant Coursework:** Applied Linear Algebra, Probability & Statistics, Design & Analysis of Algorithms, Data Structures & Algorithms, Systems Programming, Assembly Language & Machine Organization, Object-Oriented Programming

Professional Development

AI & Machine Learning Education Series by NVIDIA & Mark III Systems

February – April 2026

- Completed biweekly hands-on Jupyter workshops in ML model training (Keras, CIFAR-10), computer vision with CNNs (object detection, segmentation), and LLM fine-tuning (Hugging Face Transformers, Falcon-7B inference).

Ansible Training by Red Hat

January 2026

- Built and tested Ansible playbooks in instructor-led Red Hat labs, covering inventory management and automation of configuration and system administration tasks.